

Estimations Combined

Description

This code loads and combines the different feature value estimations (natural language writing, paleolithic signs, and proto-cuneiform) into one csv file. For example, the different names for sign sequences (e.g. “text”, “sign.expand.vec”) are now under one column “sequence”.

Session Info

Give the session info (reduced).

```
## [1] "R version 4.4.0 (2024-04-24)"  
## [1] "x86_64-pc-linux-gnu"
```

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##     filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##     intersect, setdiff, setequal, union
```

Give the package versions.

```
## [1] "1.1.4"
```

Load Data

Load csv files with quantitative feature values for sign sequences.

SIGNBASE

```

# load signbase estimations for sign sequences on objects
sign.data <- read.csv("~/Github/PaleoSigns/results/features/signBase_features.csv")
# select Aurignacian objects
sign.data <- sign.data[sign.data$archaeological_unit == "Aurignacien", ]

# select columns for direct comparison with TeDDi sample
sign.data.selected <- select(sign.data, c('object_id', 'archaeological_unit', 'coding_clean',
                                           'num.tokens', 'num.types', 'huni.chars', 'hrate.chars',
                                           'ttr.chars', 'rm.chars'))

# change column names
colnames(sign.data.selected) <- c('id', 'subcorpus', 'sequence', 'num.tokens', 'num.types',
                                   'huni.chars', 'hrate.chars', 'ttr.chars', 'rm.chars')

```

CDLI

```

# read feature values for proto-cuneiform from CDLI
# Uruk V
cuneiV.data <- read.csv("~/Github/PaleoSigns/results/features/cuneiform_UrukV_features.csv")

# add column with subcorpus information
cuneiV.data$subcorpus <- rep("Cuneiform (Uruk V)", nrow(cuneiV.data))

# select columns for direct comparison with other samples
cuneiV.data.selected <- select(cuneiV.data, c('tablet.id', 'subcorpus', 'signs.expand.vec',
                                              'num.tokens', 'num.types', 'huni.chars', 'hrate.chars',
                                              'ttr.chars', 'rm.chars'))

# rename columns
names(cuneiV.data.selected)[names(cuneiV.data.selected) == "signs.expand.vec"] <- "sequence"
names(cuneiV.data.selected)[names(cuneiV.data.selected) == "tablet.id"] <- "id"

# Uruk IV
cuneiIV.data <- read.csv("~/Github/PaleoSigns/results/features/cuneiform_UrukIV_features.csv")

# add column with subcorpus information
cuneiIV.data$subcorpus <- rep("Cuneiform (Uruk IV)", nrow(cuneiIV.data))

# select columns for direct comparison with other sample
cuneiIV.data.selected <- select(cuneiIV.data, c('tablet.id', 'subcorpus', 'signs.expand.vec',
                                              'num.tokens', 'num.types', 'huni.chars', 'hrate.chars',
                                              'ttr.chars', 'rm.chars'))

# rename column with sign sequences
names(cuneiIV.data.selected)[names(cuneiIV.data.selected) == "signs.expand.vec"] <- "sequence"
names(cuneiIV.data.selected)[names(cuneiIV.data.selected) == "tablet.id"] <- "id"

# Uruk III
cuneiIII.data <- read.csv("~/Github/PaleoSigns/results/features/cuneiform_UrukIII_features.csv")

# add column with subcorpus information
cuneiIII.data$subcorpus <- rep("Cuneiform (Uruk III)", nrow(cuneiIII.data))

# select columns for direct comparison with other sample
cuneiIII.data.selected <- select(cuneiIII.data, c('tablet.id', 'subcorpus', 'signs.expand.vec',

```

```

                                'num.tokens', 'num.types', 'huni.chars', 'hrate.chars',
                                'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(cuneiIII.data.selected)[names(cuneiIII.data.selected) == "signs.expand.vec"] <- "sequence"
names(cuneiIII.data.selected)[names(cuneiIII.data.selected) == "tablet.id"] <- "id"

```

TeDDI

```

# read feature values for character sequences of the TeDDi sample
teddi.data <- read.csv("~/Github/PaleoSigns/results/features/teddi_features.csv")
# add column with subcorpus information
teddi.data$subcorpus <- rep("TeDDi", nrow(teddi.data))
# select columns
teddi.selected <- select(teddi.data, c('id', 'subcorpus', 'text', 'num.tokens', 'num.types', 'huni.chars',
                                       'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(teddi.selected)[names(teddi.selected) == "text"] <- "sequence"

```

Combine

Combine the different data frames.

```

estimations.df <- rbind(sign.data.selected, cuneiV.data.selected,
                        cuneiIV.data.selected, cuneiIII.data.selected,
                        teddi.selected)
head(estimations.df)

```

```

##      id  subcorpus      sequence num.tokens num.types huni.chars
## 1 bst0001 Aurignacien vvvvvvvvvvvvvvvvv 18      1 0.0000000
## 2 gkl0001 Aurignacien   ||| //___/_V   12      5 1.8962406
## 3 gkl0002 Aurignacien      vvvvv      5      1 0.0000000
## 4 gkl0003 Aurignacien    vvvv vvvv    9      2 0.5032583
## 5 gkl0004 Aurignacien  vvvvvvvvvvvvvvv 14      1 0.0000000
## 6 gkl0005 Aurignacien   |||          3      1 0.0000000
##  hrate.chars ttr.chars  rm.chars
## 1  0.6171397 0.05555556 1.0000000
## 2  1.5295563 0.41666667 0.6363636
## 3  0.5711903 0.20000000 1.0000000
## 4  0.9060047 0.22222222 0.7500000
## 5  0.6379550 0.07142857 1.0000000
## 6  0.4308271 0.33333333 1.0000000

```

Simple Stats

```

nrow(estimations.df) # number of sequences with feature estimations

## [1] 3076
unique(estimations.df$subcorpus) # names of subcorpora

```

```
## [1] "Aurignacien"          "Cuneiform (Uruk V)"    "Cuneiform (Uruk IV)"
## [4] "Cuneiform (Uruk III)" "TeDDi"

nrow(estimations.df[estimations.df$subcorpus=="Aurignacien", ]) # number of Aurignacien sequences

## [1] 211

nrow(estimations.df[estimations.df$subcorpus=="TeDDi", ]) # number of TeDDi (modern writing) sequences

## [1] 1040

nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk V)", ]) # number of Uruk V proto-cuneiform sequences

## [1] 111

nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk IV)", ])

## [1] 859

nrow(estimations.df[estimations.df$subcorpus=="Cuneiform (Uruk III)", ])

## [1] 855

Write to file

write.csv(estimations.df, "~/Github/PaleoSigns/results/features/features_combined.csv", row.names = F)
```

Package References

```
## Wickham H, François R, Henry L, Müller K, Vaughan D (2023). _dplyr_: A
## Grammar of Data Manipulation_. R package version 1.1.4,
## <https://CRAN.R-project.org/package=dplyr>.
```